

# MALNUTRITION

----- Active space -----

Condition of undernutrition or overnutrition.

## TERMINOLOGY IN UNDERNUTRITION :

|             | Parameter                   | Implication                          |
|-------------|-----------------------------|--------------------------------------|
| Wasting     | Low weight for height (W/H) | Acute severe weight loss             |
| Stunting    | Low height for age          | Chronic undernutrition               |
| Underweight | Low weight for age          | Can refer to wasting and/or stunting |

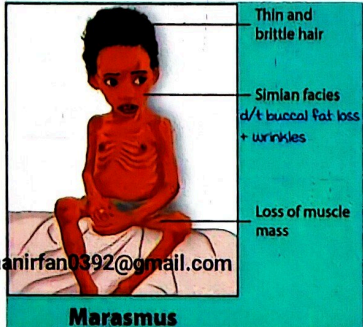
## WHO CLASSIFICATION OF MALNUTRITION :

- Based on WHO charts.

|                   | moderate malnutrition | Severe malnutrition |
|-------------------|-----------------------|---------------------|
| Wasting           | -2 to -3 S.D.         | < -3 S.D.           |
| Stunting          | -2 to -3 S.D.         | < -3 S.D.           |
| Symmetrical edema | Absent                | Present             |

## Clinical Syndromes of malnutrition

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| Protein-energy malnutrition |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | marasmus                                                                                                                                                                                                                                                                                                                                                                                                                         | Kwashiorkor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Predominant deficiency      | Calorie                                                                                                                                                                                                                                                                                                                                                                                                                          | Protein                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Onset                       | <1 yr                                                                                                                                                                                                                                                                                                                                                                                                                            | 1-4 yrs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Edema                       | Absent                                                                                                                                                                                                                                                                                                                                                                                                                           | Present (+++); Anasarca                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| muscle wasting              | Present (+++)                                                                                                                                                                                                                                                                                                                                                                                                                    | Present (+), but masked by edema                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Appetite                    | Good                                                                                                                                                                                                                                                                                                                                                                                                                             | Poor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| General appearance          | <ul style="list-style-type: none"> <li>Alert.</li> <li>Skin and bone appearance.</li> <li>Loose skin folds in axilla &amp; buttocks : (Baggy pant appearance).</li> </ul>  <p>Thin and brittle hair</p> <p>Similar facies d/t buccal fat loss + wrinkles</p> <p>Loss of muscle mass</p> <p>khanirfan0392@gmail.com</p> <p><b>Marasmus</b></p> | <ul style="list-style-type: none"> <li>Dull/lethargic/irritable.</li> <li>Skin changes : Flaky-paint appearance; Hyperpigmented patches.</li> <li>Flag sign (Alternating grey &amp; black patches in the hair).</li> <li>↓ S. Albumin.</li> <li>↓ oncotic pressure</li> <li>Anasarca</li> </ul>  <p>Thin and brittle hair</p> <p>Moon face</p> <p>Bulging abdomen (ascites)</p> <p><b>Kwashiorkor</b></p> <p>Crazy pavement appearance. (Cracks on skin).</p> |

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Nutrition

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**Severe Acute Malnutrition (SAM)**

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**CRITERIA :**

malnourished Child

1. Weight for Height :  
< -3.5 SD.2. Bipedal edema :  
Diagnosis of exclusion3. mean Arm Circumference  
(MAC) : < 11.5 cm

- Presence of any of the above : SAM.

**TYPES OF SAM :**

|                       | Uncomplicated SAM          | Complicated SAM     |
|-----------------------|----------------------------|---------------------|
| Appetite              | Good                       | Poor                |
| Generalized edema     | Absent                     | Present             |
| Medical Complications | Absent                     | Present             |
| Intervention          | Supervised Home management | Hospital management |

**HOME MANAGEMENT OF SAM :**

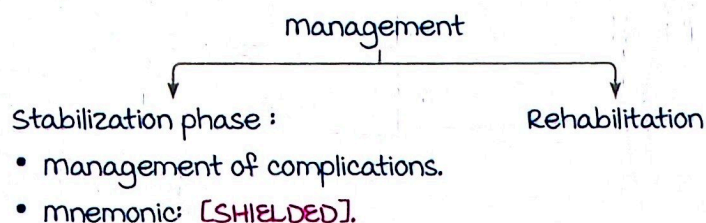
Nutrition goal : 175 Kcal/kg/day + Protein: 4-6 g/kg/day.

**Diets:**

- Home-made foods.
- **Ready-to-use therapeutic food (RUTF).**
- Ingredients :
  1. milk Solids.
  2. Sugars.
  3. Vegetable oil.
  4. Peanut butter & paste.
- Advantages :
  1. Good palatability.
  2. Semi-solid consistency.
  3. Long shelf-life.
- Nutrient Value :  
543 Kcal + 15g protein  
(Per 100g).

**Additional Interventions (3 As) :**

- Antibiotic : Oral Amoxicillin x 5d.
- Albendazole (Deworming) : Dosage
  - <2y : 200mg
  - >2y : 400mg
- Vitamin A : Dosage :
  - <6 months - 50,000 IU
  - 6-12 months - 1 Lakh IU
  - >12 months - 2 Lakh IU

**HOSPITAL MANAGEMENT OF SAM :**

## Phase I : Stabilization

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| Complications                   | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Hypoglycemia (Sugars)        | <ul style="list-style-type: none"> <li>Definition : Blood glucose : <math>&lt; 54</math> mg/dl.</li> <li>Treatment :               <div style="text-align: center;">                 Hypoglycemia<br/>                 ↓<br/> <div style="display: flex; justify-content: space-around;"> <div>                     Symptomatic (Eg : Seizures)<br/>                     IV 10% Dextrose : 5ml/kg STAT<br/>                     ↓<br/>                     Glucose infusion                 </div> <div>                     Asymptomatic<br/>                     Oral glucose : 50mL                 </div> </div> </div> </li> </ul>                                     |
| 2. Hypothermia                  | <ul style="list-style-type: none"> <li>Definition : Axillary temp : <math>&lt; 35^{\circ}\text{C}</math> / Rectal temperature : <math>&lt; 35.5^{\circ}\text{C}</math>.</li> <li>Treatment :               <ul style="list-style-type: none"> <li>- Warming child using protective clothing.</li> <li>- Kangaroo mother care in infants.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                             |
| 3. Infections                   | <p style="text-align: center;">↓ immunity<br/>(usual signs may be absent)</p> <p style="text-align: center;">↓</p> <p style="text-align: center;">Empirical treatment is started in all children of SAM.</p> <p style="text-align: center;">m/c antibiotics : Ampicillin + Amikacin (Penicillin + Aminoglycoside)</p> <p style="text-align: center;">↓ If no improvement in 48hrs.</p> <p style="text-align: center;">3rd generation cephalosporins (Eg : Ceftriaxone).</p> <p style="text-align: center;">m/c bacteria : Gram-negative.</p>                                                                                                                                |
| 4. Electrolyte imbalance        | <ul style="list-style-type: none"> <li>Sodium levels : ↑ total body overall (Excess <math>\text{Na}^+</math> avoided).               <ul style="list-style-type: none"> <li>- Serum sodium : Low/normal: (D/t intracellular sodium &amp; dilution effect of edema).</li> </ul> </li> <li>Hypokalemia</li> <li>Hypomagnesemia</li> </ul> <p style="text-align: right;">} Treatment : Supplementation</p>                                                                                                                                                                                                                                                                     |
| 5. Dehydration (DE)             | <p>Clinical Symptoms :</p> <ul style="list-style-type: none"> <li>• Thirst.</li> <li>• Urine output.</li> <li>• Dryness of oral mucosa.</li> </ul> <p>Treatment : Started irrespective of symptoms.</p> <div style="display: flex; justify-content: space-around;"> <div> <p>No shock symptoms :</p> <p>Oral fluids : Rehydration Solution for malnutrition (Resomal).</p> </div> <div> <p>Shock symptoms present :</p> <ul style="list-style-type: none"> <li>• Ringer's lactate with 5% dextrose.</li> <li>• Contains <math>\text{K}^+</math> ions.</li> </ul> <p style="text-align: center;">First 2 hours : 15 ml/kg<br/>↓ Resolution<br/>Oral fluids</p> </div> </div> |
| 6. micronutrient Deficiency (D) | <ul style="list-style-type: none"> <li>multivitamin supplement recommended for all SAM.               <ul style="list-style-type: none"> <li>- Dose : Twice RDA value.</li> </ul> </li> <li>Iron absorption in first week ↓.               <div style="text-align: center;">                 ↓<br/>                 unabsorbed iron               </div> <div style="display: flex; align-items: center;"> <div style="font-size: 2em; margin-right: 10px;">{</div> <div>                     Free radical damage<br/>                     Promote bacterial proliferation                 </div> </div> </li> </ul>                                                        |

Note :

Rapid rewarming should be avoided as it causes dysequilibrium syndrome.

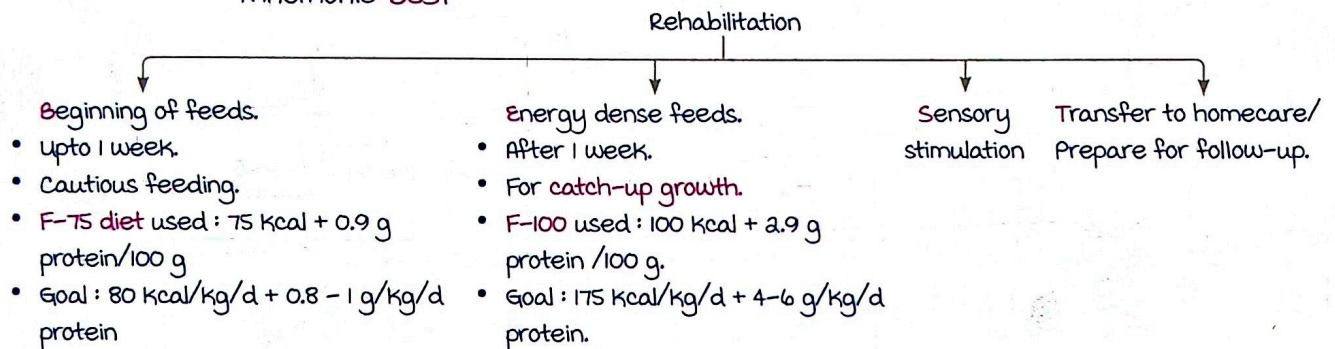
- Causes : Drowsiness, seizures, coma.

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- Hypoglycemia, hypothermia, & infection usually occur as a **triad**: Treatment initiated for all three if one is present.
- Skin pinch test is an unreliable indicator of dehydration as there is low subcutaneous fat.

**ReSOMAL v/s WHO ORS Solution :**

|                     | WHO ORS | ReSOMAL |
|---------------------|---------|---------|
| Osmolarity (mOsm/L) | 245     | 300     |
| Sodium (Na)         | 75      | 45      |
| Glucose             | 75      | 125     |
| Potassium (K)       | 20      | 40      |
| Chloride (Cl)       | 65      | 70      |

**Phase II: Rehabilitation :**mnemonic: **BEST**

|                       | 648c85cfee3b03a74e182fab<br>Stabilization |          | Rehabilitation |
|-----------------------|-------------------------------------------|----------|----------------|
|                       | Day 1-2                                   | Days 3-7 | Weeks 2-6      |
| Hypoglycemia          | ----->                                    |          |                |
| Hypothermia           | ----->                                    |          |                |
| Dehydration           | ----->                                    |          |                |
| Electrolytes          | ----->                                    |          |                |
| Infection             | ----->                                    |          |                |
| Micronutrients        | No iron                                   |          | With iron      |
| Cautious feeding      | ----->                                    |          |                |
| Catch-up growth       |                                           |          | ----->         |
| Sensory stimulation   | ----->                                    |          |                |
| Prepare for follow-up |                                           |          | ----->         |

**Summary of management****Treatment completion :**

No edema for at least 2 weeks

+

Weight for height : **-2 SD/MAC** reaches 12.5 cmNote : Deworming to be done prior to discharge - **Albendazole 400mg** single dose.